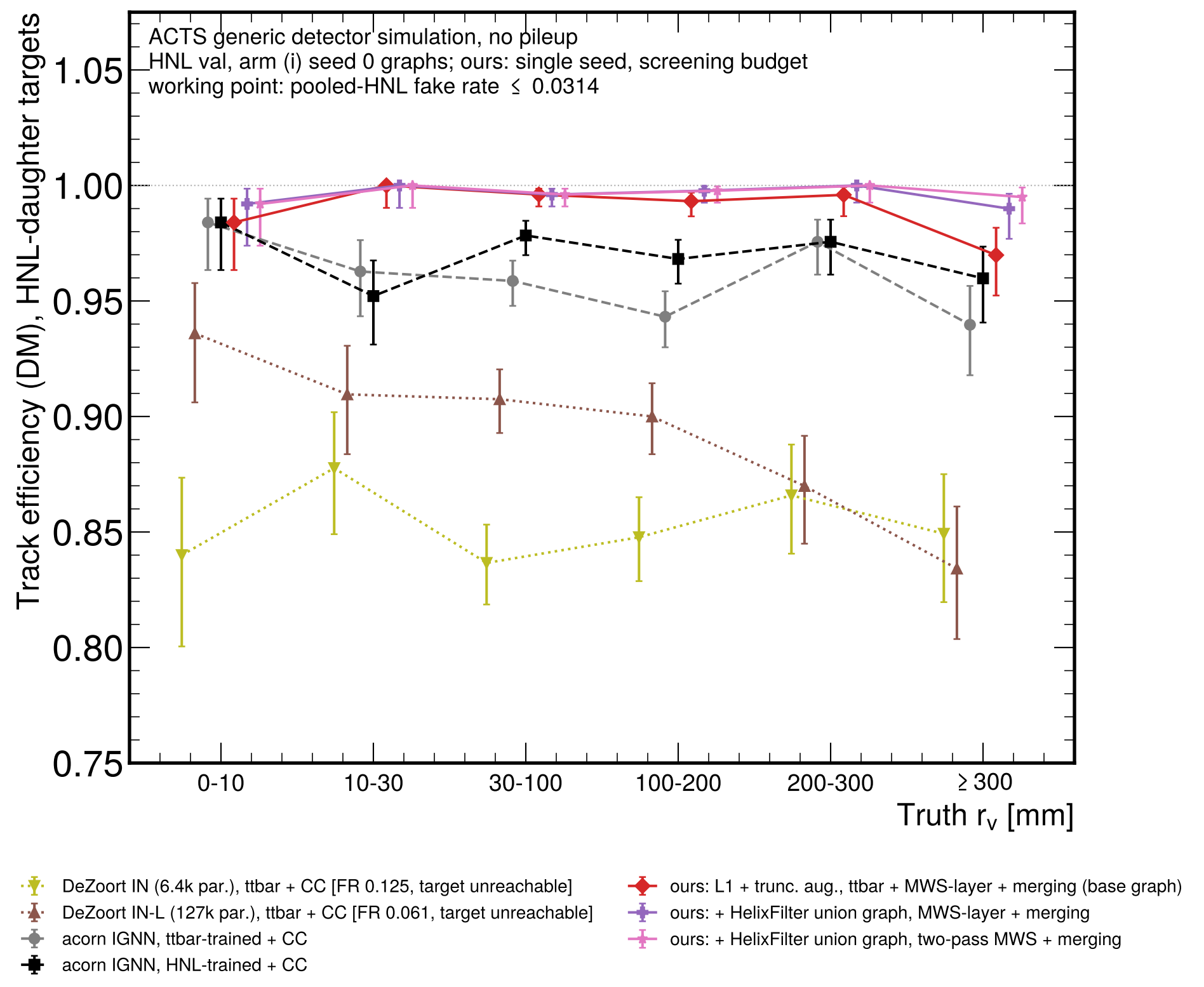
ACTS generic detector simulation, no pileup
HNL val, arm (i) seed 0 graphs; ours: single seed, screening budget
working point: pooled-HNL fake rate ≤ 0.0314
Track efficiency (DM), HNL-daughter targets
1.05
1.00
0.95
0.90
0.85
0.80
0.75
0-10
10-30
30-100
100-200
200-300
≥ 300
Truth r_v [mm]
DeZoort IN (6.4k par.), ttbar + CC [FR 0.125, target unreachable]
DeZoort IN-L (127k par.), ttbar + CC [FR 0.061, target unreachable]
acorn IGNN, ttbar-trained + CC
acorn IGNN, HNL-trained + CC
ours: L1 + trunc. aug., ttbar + MWS-layer + merging (base graph)
ours: + HelixFilter union graph, MWS-layer + merging
ours: + HelixFilter union graph, two-pass MWS + merging